

OTP

OTP (One Time Programmable) is a kind of non-volatile memory technology, which can only be programmed once. Once the memory has been programmed, it cannot be changed or reprogrammed, but it can still retain data after power failure. OTP memory is usually used to store critical data, such as security keys or system configuration information, to prevent unauthorized access or tampering.

The default value of each bit in RTS3917 and RTS3918 OTP is 1, and it can be modified to 0. The entire OTP is divided into 57 Groups. The specific details are as follows:

- Group 0 has 128 bits, and each bit has a specific function.
- For Group 1 to Group 56, each Group contains 288 bits. Among them, the lower 256 bits are the data segment, and the higher 32 bits are the parity bits (ECC).

The allocation of Groups is shown in the table1 below

Table1: Group allocation

Group	Application
Group0	Function BIT, see the detailed description of the functions of each BIT in Group0 in the table below.
Group1-4	4 groups of keys for use by the AES hardware module.
Group5-6	2 groups of SHA256 values storing the RSA public keys.
Group7-10	4 groups of stored Initialization IV.
Group11	Ethernet Mac
Group12-56	Dummy Group

The function descriptions of each BIT in Group0 are shown in the following table. "Function descriptions of each BIT in Group0"

Table:Function descriptions of each BIT in Group0

Group	BIT	description
AES Group	0-3	Control the programming and reading permissions of Group1 to Group4: 1: The group is programmable and the CPU can read it. 0: The group is not programmable and the CPU cannot read it.
RSA Group	4-5	Control the programming and reading permissions of Group5 to Group6: 1: The group is programmable and the CPU can read it. 0: The group is not programmable, but the CPU can read it.
IV Group	6-9	Control the programming and reading permissions of Group7 to Group10: 1: The group is programmable and the CPU can read it. 0: The group is not programmable, but the CPU can read it.
Other Group	10-56	Control the programming and reading permissions of Group11 to Group57: 0: The group is not programmable, but the CPU can read it. 1: The group is programmable and the CPU can read it.
IC Information Flag	95	Encryption boot flag bits: 1: Read the key from the SD card. 0: Read the key from the One-Time Programmable (OTP) memory.
	97	Secure boot flag bits: 1: Non-secure boot. 0: Secure boot.

Warning: Once each bit in the One-Time Programmable (OTP) is programmed, it cannot be changed. Therefore, extreme caution must be exercised when programming the OTP.

Once the data is written into the One-Time Programmable (OTP), it cannot be changed again. In particular, for Group1 where the AES key is located, after the writing is completed and the control bits are set to 0, the tool will not be able to read the actual value of this Group and will always display all zeros. Therefore, users need to take good care of all the written keys.

You can use "otp_mfg -r -pg" to check whether the writing is correct.

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# otp_mfg -r -pg
group000-000(0000): ae ff ff ff ff ff ff ff ff ff ff 7f fd ff ff ff
group001-002(0016): 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 // aes-key
group003-004(0052): ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff
group005-006(0088): ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff
group007-008(0124): ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff
group009-010(0160): d2 29 57 af 9d 06 54 87 9c 7d 6d 3a 8b 53 94 99 46 82 db 81 74 // Hash of ROTPK
group011-012(0196): ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff ff
group013-014(0232): 62 e6 f2 98 44 86 d8 f7 ea 05 9a ca 8b 67 af 3b ff ff ff ff ff // aes-iv
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